Keyush Shah

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EDUCATION

University of Pennsylvania (School of Engineering and Applied Science) **Philadelphia, PA**
*Master of Science in Engineering (**MSE**) in Data Science; **GPA 3.9/4.0*** ***Aug 2023 – May 2025***
Coursework: *Machine Learning, Databases, Computer Systems, Deep Learning, Generative Modeling, NLP, Machine Perception*

EXPERIENCE

Machine Learning Engineer, Omnicom (NYC, New York) ***Aug 2025 - Present***

- Designed a RAG based context-augmented LLM chat system in our healthcare analytics SaaS product with **5+ API endpoints and 8 MongoDB collections** helping clients simulate patient focus groups through survey-based insights.
- Led end-to-end migration of a multi-component ML Pipeline from AzureML to Databricks by re-architecting LLM Integration, data storage and maintaining orchestration in Spark Job DAGs unifying pipeline governance.
- Developed CPU-only **BERTopic** pipeline, replacing the GPU compute node & fetching same output at **~90% lower cost**.
- Reduced CI/CD pipelines runtime by **12x** by migrating from Pipenv to **UV package manager**, configuring private package indexes and benchmarking Docker containers - packages installation slashed to 12s across backend services.
- Centralized embedding generation across Databricks pipelines by replacing OpenAI SDK instantiation with endpoint wrapper reducing **~1000 per-run API calls by 10x** through request batching & eliminating per-call auth overhead.

AI Researcher, *Penn Medicine* - *Computational Social Listening Lab* (PA, USA) ***Dec 2024 – Present***
IH Risk Model [CODE LINK]

- Reduced **model tuning time by 10x** by developing an **AutoML** pipeline using sklearn for **hernia** prediction **(recall: 0.9)**
- Deployed GPT-4 few-shot pipeline on 10k+ redacted surgical notes, **reducing 30% noise vs. BERT** for feature labeling.
- Built a fault tolerant multi-agent orchestration system using Claude Agent SDK featuring **LLM-as-judge** observability, automated retry logic and **FastAPI + WebSocket web interface** slashing pipeline runtime from **45 to 10 min.**
- Integrated **vLLM** & **LangChain** for scalable local LLM inference deployed on A100 GPU and **modularized the backend** (OpenAI, Mistral) using **parallelized batch inference** and **memory-aware chunking** across variable length notes.

Data Science Intern (Full-time), ***Universal Media*** (PA, USA) ***May 2024 – Aug 2024***

- Led the development of **3+ data pipelines** using Azure Data Factory, facilitating seamless ingesting into Azure
- Developed python scripts for data transformation, stored them in Blob storage and executed them via batch activity in ADF.
- Drove product insights by building **Mixed Media time series models in Azure Synapse**, analyzing marketing channel impacts on media diversity metrics. Built **Power BI** dashboards to deliver actionable insights for optimizing client strategies.

Assistant Manager (Full-time), ***IIFL Finance Ltd*** ***Apr 2022 – July 2023***

- Analyzed ETL process failures and created **10+** paginated reports in SSIS to help the management track 1000+ branches.
- **Digital Adoption:** Led a product-focused initiative to identify digitally savvy customers by engineering features and building ADF pipelines to track campaign behavior. Trained and deployed a Random Forest model (**with a 90% accuracy**) in Azure ML Studio; **exposed it as a REST endpoint consumed by marketing campaigns,** driving digital **disbursal adoption by 50%.**

SELECTED PROJECTS

- **Particle (2024):**
  - Built a context-aware chatbot for product and order support using FastAPI, LangChain, and Pinecone, implementing **RAG for product queries and tool-based logic with LLM fallback** for transactions.
  - Engineered multi-turn memory and entity tracking, **enabling >95% accurate follow-up handling**. Automated ingestion of synthetic data, & delivered **React + Vite frontend, improving support efficiency by 40%** and user satisfaction. [Link]
- **Ride Duration Prediction (2025):** Developed a production-ready ML pipeline to predict NYC taxi ride durations, using **Airflow** for orchestration and **MLflow** for experiment tracking. Achieved **~30% RMSE reduction** via automated hyperparameter tuning & designed modular, reproducible workflows to simulate real-world deployment as a web service via **Flask**. [Link]
- **Multithreaded Image Processing (2025):** Engineered a parallelized box blur algorithm in C++ using POSIX threads, achieving a 2.8× speedup (3251 ms → 1165 ms) on 4 cores by optimizing memory access, leveraging shared-memory synchronization, and **partitioning workloads across non-overlapping thread-local regions**. [Link]

TECHNICAL SKILLS

Programming Languages: Python, C/C++, SQL, JavaScript, Typescript, R programming
ML Libraries/Frameworks: PyTorch, scikit-learn, XGBoost, LightGBM, HuggingFace, spaCy, OpenCV, MLflow, A/B Testing
Databases/Web Frameworks: MySQL, PostgreSQL, SSMS, MongoDB, Neo4j, React, NodeJS, Django, Flask, FastAPI
Cloud/Big Data Orchestration: AWS (S3, Glue), Azure (Data Factory, Synapse), GCP (BigQuery), Kafka, Airflow, DBT, PySpark
Tools/DevOps: DataBricks, Apache Spark, Docker, Jenkins, Git, Kubernetes, pytest, CI/CD